1)
$|\mathbf{A}| = ad - bc$

2) a)
$\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 4 = 6 - 4 = 2$

2) b)
$\begin{vmatrix} 6 & -4 \\ 3 & 5 \end{vmatrix} = 6 \times 5 - (-4) \times 3 = 30 + 12 = 42$

2) c)
$\begin{vmatrix} 3 & 2 \\ 8 & -5 \end{vmatrix} = 3 \times (-5) - 2 \times 8 = -15 - 16 = -31$

2) d)
$\begin{vmatrix} -6 & -5 \\ 3 & 2 \end{vmatrix} = -6 \times 2 - (-5) \times 3 = -12 + 15 = 3$

3) a)
$\begin{vmatrix} k & 5 \\ 2k & 3 \end{vmatrix} = k \times 3 - 5 \times 2k = 3k - 10k = -7k$

3) b)
$\begin{vmatrix} k & 4 \\ 5 & 3k \end{vmatrix} = k \times 3k - 4 \times 5 = 3k^2 - 20$

3) c)
$\begin{vmatrix} 3k & k+1 \\ 4 & k-2 \end{vmatrix} = 3k \times (k-2) - (k+1) \times 4 = 3k(k-2) - 4(k+1)$
$= 3k^2 - 6k - 4k - 4 = 3k^2 - 10k - 4$

3) d)
$\begin{vmatrix} k-2 & k-3 \\ k & 2k+1 \end{vmatrix} = (k-2) \times (2k+1) - (k-3) \times k = (k-2)(2k+1) - (k-3)k$
$= 2k^2 - 3k - 2 - k^2 + 3k = k^2 - 2$

4) a)
$\begin{vmatrix} p & q \\ q & p \end{vmatrix} = p \times p - q \times q = p^2 - q^2$

4) b)

$$\begin{vmatrix} p & p-3 \\ q & q+2 \end{vmatrix} = p \times (q+2) - (p-3) \times q = pq + 2p - pq + 3q = 2p + 3q$$

4) c)

$$\begin{vmatrix} q-1 & p-1 \\ p+1 & q+1 \end{vmatrix} = (q-1) \times (q+1) - (p-1) \times (p+1)$$
$$= (q^2 - 1) - (p^2 - 1) = q^2 - p^2$$

4) d)

$$\begin{vmatrix} q+2 & p+3 \\ p-4 & q-3 \end{vmatrix} = (q+2) \times (q-3) - (p+3) \times (p-4)$$
$$= (q^2 - q - 6) - (p^2 - p - 12)$$
$$= q^2 - q - p^2 + p + 6$$

5)

$$\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \cos\theta \times \cos\theta - (-\sin\theta) \times \sin\theta = \cos^2\theta + \sin^2\theta = 1$$

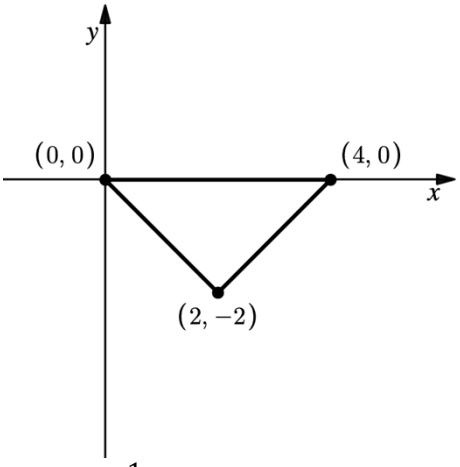
6) a)

$$\begin{vmatrix} \ln 8 & \ln 2 \\ \ln 4 & \ln 16 \end{vmatrix} = \begin{vmatrix} \ln 2^3 & \ln 2 \\ \ln 2^2 & \ln 2^4 \end{vmatrix} = \begin{vmatrix} 3\ln 2 & \ln 2 \\ 2\ln 2 & 4\ln 2 \end{vmatrix}$$
$$= (3\ln 2) \times (4\ln 2) - (\ln 2) \times (2\ln 2)$$
$$= 12(\ln 2)^2 - 2(\ln 2)^2$$
$$= 10(\ln 2)^2$$

6) b)

$$\begin{vmatrix} \ln 81 & \ln 27 \\ \ln 9 & \ln 3 \end{vmatrix} = \begin{vmatrix} \ln 3^4 & \ln 3^3 \\ \ln 3^2 & \ln 3 \end{vmatrix} = \begin{vmatrix} 4\ln 3 & 3\ln 3 \\ 2\ln 3 & \ln 3 \end{vmatrix}$$
$$= (4\ln 3) \times (\ln 3) - (3\ln 3) \times (2\ln 3)$$
$$= 4(\ln 3)^2 - 6(\ln 3)^2$$
$$= -2(\ln 3)^2$$

7) a)



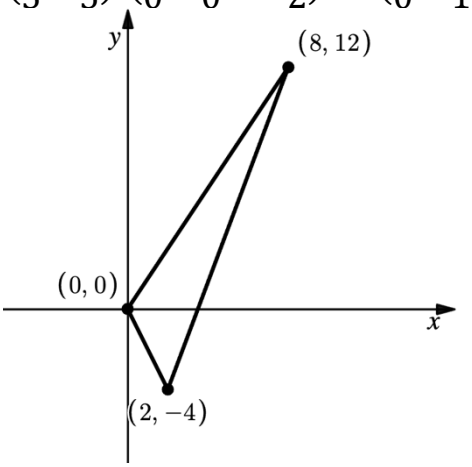
Area $= \frac{1}{2} \times 4 \times 2 = 4$ units^2

7) b)

$\det \mathbf{A} = 2 \times 5 - 1 \times 3 = 10 - 3 = 7$

7) c)

$$\begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 0 & 4 & 2 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 8 & 2 \\ 0 & 12 & -4 \end{pmatrix}$$



Area $= 16 \times 8 - \frac{1}{2} \times 8 \times 12 - \frac{1}{2} \times 2 \times 4 - \frac{1}{2} \times 6 \times 16 = 28$ units^2

7) d)

The determinant is the area scale factor

i.e. 4 units$^2 \times 7 = 28$ units^2

8)

The area of Q is $6 \times 15 = 90$ units^2

9)

The smallest possible area of D is $3150 \times 24.5 = 77175$ units^2

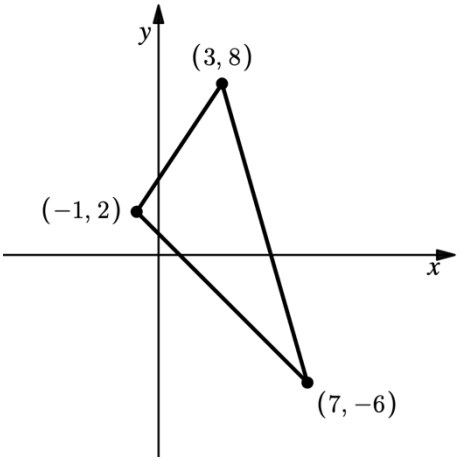
10)

$\begin{vmatrix} 5 & 8 \\ 2 & 6 \end{vmatrix} = 5 \times 6 - 8 \times 2 = 30 - 16 = 14$

5 cm$^2 \times 14 = 70$ cm^2

So the area of S' is 70 cm^2

11) a)



Area $= 8 \times 14 - \frac{1}{2} \times 4 \times 6 - \frac{1}{2} \times 8 \times 8 - \frac{1}{2} \times 4 \times 14 = 40$ units^2

11) b)

$\det \mathbf{A} = 3 \times 6 - (-2) \times 4 = 18 + 8 = 26$

11) c)

40 units$^2 \times 26 = 1040$ units^2

12) a)

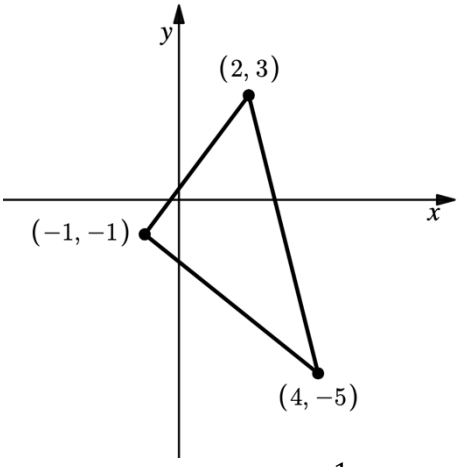
$\begin{pmatrix} 5-k & 3 \\ -4 & 4 \end{pmatrix} \begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix} = \begin{pmatrix} 19-2k & 5-4k & k-8 \\ 4 & -36 & 0 \end{pmatrix}$

So the coordinates of the vertices of T' are $(19-2k, 4)$, $(5-4k, -36)$ and $(k-8, 0)$

12) b)

$\det \mathbf{A} = (5-k) \times 4 - 3 \times (-4) = 4(5-k) + 12 = 20 - 4k + 12 = 32 - 4k$

12) c)



Area of $T = 5 \times 8 - \frac{1}{2} \times 3 \times 4 - \frac{1}{2} \times 5 \times 4 - \frac{1}{2} \times 2 \times 8 = 16$ units^2

16 units$^2 \times (32 - 4k) = 320$ units^2

$16(32 - 4k) = 320 \Rightarrow 32 - 4k = 20 \Rightarrow 4k = 12 \Rightarrow k = 3$

13) a)

$$\begin{pmatrix} 4-k & 2 \\ -6 & k+3 \end{pmatrix}\begin{pmatrix} -4 & 1 & 2 \\ 1 & 6 & -7 \end{pmatrix} = \begin{pmatrix} 4k-14 & 16-k & -2k-6 \\ k+27 & 6k+12 & -7k-33 \end{pmatrix}$$

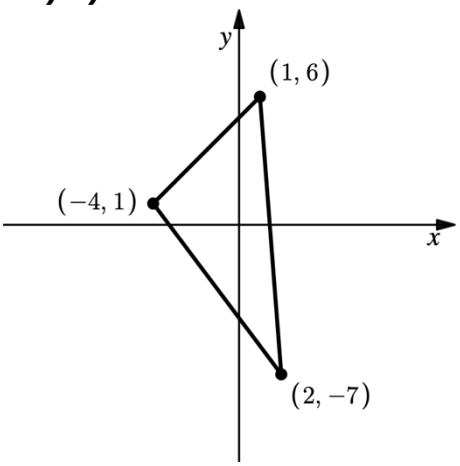
So the coordinates of the vertices of T' are $(4k - 14, k + 27)$, $(16 - k, 6k + 12)$ and $(-2k - 6, -7k - 33)$

13) b)

$\det \mathbf{A} = (4 - k) \times (k + 3) - 2 \times (-6) = (4 - k)(k + 3) + 12 = -k^2 + k + 12 + 12$

$= -k^2 + k + 24$

13) c)



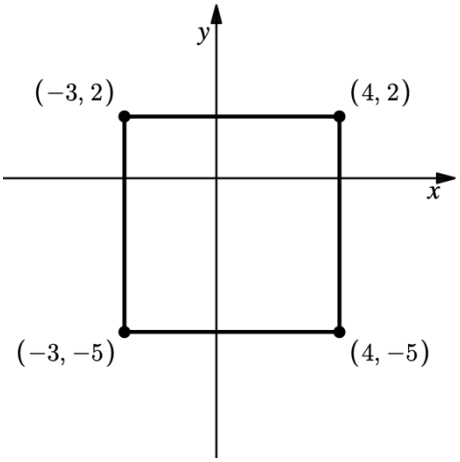
Area of $T = 13 \times 6 - \frac{1}{2} \times 5 \times 5 - \frac{1}{2} \times 6 \times 8 - \frac{1}{2} \times 1 \times 13 = 35$ units^2

35 units$^2 \times (-k^2 + k + 24) = 140$ units^2

$35(-k^2 + k + 24) = 140 \Rightarrow -k^2 + k + 24 = 4 \Rightarrow k^2 - k - 20 = 0$

$\Rightarrow (k - 5)(k + 4) = 0 \Rightarrow k = 5$ only as k is a positive constant

14)



Area of $S = 7 \times 7 = 49$ units^2
$\det \mathbf{A} = (2k+1) \times k - 5k \times 3 = (2k+1)k - 15k = 2k^2 + k - 15k = 2k^2 - 14k$
49 units$^2 \times (2k^2 - 14k) = 784$ units^2
$49(2k^2 - 14k) = 784 \Rightarrow k^2 - 7k = 8 \Rightarrow k^2 - 7k - 8 = 0 \Rightarrow (k-8)(k+1)$
$\Rightarrow k = -1$ or $k = 8$

15)
$\begin{vmatrix} 60 & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{vmatrix} = 60 \times \frac{1}{2\sqrt{2}} - 0 \times 0 = 15\sqrt{2}$
Area of $T = \frac{1}{2} \times 0.025 \times \left(\sqrt{2} + \sqrt{8}\right) = \frac{3\sqrt{2}}{80}$ cm^2
Area of $T' = 15\sqrt{2} \times \frac{3\sqrt{2}}{80} = \frac{9}{8} = 1.125$ cm^2

16) a)
$\det \mathbf{A} = 6 \times 1 - 2 \times 2 = 6 - 4 = 2$
Area of $T' = 8 \times 2 = 16$ units^2

16) b)
$\det \mathbf{C} = 6 \times 3 - (-2) \times 1 = 18 + 2 = 20$
Area of $T' = 8 \times 20 = 160$ units^2

16) c)
$\det \mathbf{B} = (-2) \times (-3) - 4 \times 2 = 6 - 8 = -2$
Area of $T' = 8 \times 2 \times |-2| = 32$ units^2

16) d)
Area of $T' = 8 \times |-2| \times 2 = 32$ units^2

16) e)
Area of $T' = 8 \times 2 \times 20 = 320$ units^2

16) f)
Area of $T' = 8 \times |-2| \times 20 = 320$ units^2

16) g)
Area of $T' = 8 \times 2 \times |-2| \times 20 = 640$ units^2

17) a)
$k \times 1 - 3 \times 5 = 3 \Rightarrow k - 15 = 3 \Rightarrow k = 18$

17) b)
$k \times 4 - 8 \times 5 = 16 \Rightarrow 4k - 40 = 16 \Rightarrow 4k = 56 \Rightarrow k = 14$

17) c)
$k \times k - 3 \times 4 = 4 \Rightarrow k^2 - 12 = 4 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$

17) d)
$k \times 9 - 5 \times k = 40 \Rightarrow 9k - 5k = 40 \Rightarrow 4k = 40 \Rightarrow k = 10$

17) e)
$k \times 1 - k \times 3 = -6 \Rightarrow k - 3k = -6 \Rightarrow -2k = -6 \Rightarrow k = 3$

17) f)
$k \times 10 - k \times k = 9 \Rightarrow 10k - k^2 = 9 \Rightarrow 0 = k^2 - 10k + 9$
$\Rightarrow 0 = (k - 9)(k - 1) \Rightarrow k = 9$ or $k = 1$

17) g)
$2k \times k - k \times 4 = 2 \Rightarrow 2k^2 - 4k = 2 \Rightarrow k^2 - 2k - 1 = 0 \Rightarrow k = 1 \pm \sqrt{2}$

17) h)
$3k \times 4k - 6 \times k = 6 \Rightarrow 12k^2 - 6k = 6 \Rightarrow 2k^2 - k - 1 = 0$
$\Rightarrow (2k + 1)(k - 1) = 0 \Rightarrow k = -\frac{1}{2}$ or $k = 1$

17) i)
$(k + 1) \times k - 3 \times 4 = 8 \Rightarrow k^2 + k - 12 = 8 \Rightarrow k^2 + k - 20 = 0$
$\Rightarrow (k + 5)(k - 4) = 0 \Rightarrow k = -5$ or $k = 4$

17) j)

$(2k-1)\times(k-2)-4\times5=7\Rightarrow 2k^2-5k+2-20=7\Rightarrow 2k^2-5k-25=0$

$\Rightarrow(2k+5)(k-5)=0\Rightarrow k=-\frac{5}{2}$ or $k=5$

17) k)

$k\times6-(k-4)\times(2k+5)=-6\Rightarrow 6k-(2k^2-3k-20)=-6$

$\Rightarrow 6k-2k^2+3k+20=-6\Rightarrow 0=2k^2-9k-26$

$\Rightarrow(2k-13)(k+2)=0\Rightarrow k=\frac{13}{2}$ or $k=-2$

17) l)

$3\times7k-(5k+1)\times(8-k)=15\Rightarrow 21k-(-5k^2+39k+8)=15$

$\Rightarrow 21k+5k^2-39k-8=15\Rightarrow 5k^2-18k-23=0$

$\Rightarrow(5k-23)(k+1)=0\Rightarrow k=\frac{23}{5}$ or $k=-1$

18)

$\det\mathbf{A}>0\Rightarrow 2\times8-k\times3>0\Rightarrow 16-3k>0\Rightarrow -3k>-16\Rightarrow k<\frac{16}{3}$

19)

$\det\mathbf{A}<0\Rightarrow 3k\times(-6)-(-4)\times2<0\Rightarrow -18k+8<0\Rightarrow -18k<-8\Rightarrow k>\frac{4}{9}$

20)

$\det\mathbf{A}>0\Rightarrow k\times(4k-4)-(k+2)\times2>0\Rightarrow 4k^2-4k-(2k+4)>0$

$\Rightarrow 4k^2-4k-2k-4>0\Rightarrow 4k^2-6k-4>0\Rightarrow 2k^2-3k-2>0$

$\Rightarrow(2k+1)(k-2)>0\Rightarrow k<-\frac{1}{2}$ or $k>2$

21)

$\det\mathbf{A}<0\Rightarrow(k-1)\times(2k+1)-(1-k)\times k<0\Rightarrow 2k^2-k-1-(k-k^2)<0$

$\Rightarrow 2k^2-k-1-k+k^2<0\Rightarrow 3k^2-2k-1<0$

$\Rightarrow(3k+1)(k-1)<0\Rightarrow -\frac{1}{3}<k<1$

22)

$\det\mathbf{A}=(2+\mathrm{i})\times\mathrm{i}-3\times(-2)=2\mathrm{i}+\mathrm{i}^2+6=2\mathrm{i}-1+6=5+2\mathrm{i}$

23) a)

Rotations maintain orientation, so matrices that represent rotations will have a positive determinant.

23) b)
Reflections reverse orientation, so matrices that represent reflections will have a negative determinant.

23) c)
Enlargements maintain orientation, so matrices that represent enlargements will have a positive determinant.

24) a)
$k = \pm 1$

24) b)
$k > 0$

25) a)
$\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = 1 \times (-1) - 0 \times 0 = -1 < 0$
As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25) b)
$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times 1 = -1 < 0$
As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25) c)
$\begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = (-1) \times (-1) - 0 \times 0 = 1 > 0$
As the determinant is positive, the matrix maintains orientation, and so does not represent a reflection.

25) d)
$\begin{vmatrix} 0 & -1 \\ 0 & 1 \end{vmatrix} = 0 \times 1 - (-1) \times 0 = 0$
As the determinant is zero, all points are mapped to the origin, and so does not represent a reflection.

25) e)

$\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = (-1) \times 1 - 0 \times 0 = -1 < 0$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

25) f)

$\begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = 0 \times 0 - (-1) \times (-1) = -1 < 0$

As the determinant is negative, the matrix reverses orientation, and so represents a reflection.

26) a)

$\begin{vmatrix} 4 & -6 \\ -9 & 3 \end{vmatrix} = 4 \times 3 - (-6) \times (-9) = 12 - 54 = -42 < 0$

As the determinant is negative, the matrix reverses orientation.

26) b)

$\begin{vmatrix} -5 & 7 \\ 1 & -2 \end{vmatrix} = (-5) \times (-2) - 7 \times 1 = 10 - 7 = 3 > 0$

As the determinant is positive, the matrix maintains orientation.

26) c)

$\begin{vmatrix} 8 & -9 \\ 10 & -3 \end{vmatrix} = 8 \times (-3) - (-9) \times 10 = -24 + 90 = 66 > 0$

As the determinant is positive, the matrix maintains orientation.

26) d)

$\begin{vmatrix} 21 & 10 \\ 15 & 7 \end{vmatrix} = 21 \times 7 - 10 \times 15 = 147 - 150 = -3 < 0$

As the determinant is negative, the matrix reverses orientation.

27) a)

Area $= 10 \times 18 = 180$ units^2

27) b)

A', D', C', B'

28)
As $k^2 \geq 0$ for all real values of k, we must have that $4k^2 + 2 \geq 2$ for all real values of k. So $\det \mathbf{A} = 4k^2 + 2 > 0$ for all real values of k and hence cannot reverse orientiation.

29)
The transformation will reverse orientation if $\det \mathbf{A} = 9 - 4k^2 < 0$
$\Rightarrow (3 - 2k)(3 + 2k) < 0 \Rightarrow k < -\frac{3}{2}$ or $k > \frac{3}{2}$
i.e. $k \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right)$

30)
$|\mathbf{A}| = 80 \times 2 - (3k) \times 9 = 160 - 27k$
The transformation will reverse orientation if $\det \mathbf{A} = 160 - 27k < 0$
$\Rightarrow k > \frac{160}{27} = 5.9259 \ldots$
As k is a positive integer, the smallest value of k will be 6

31) a)
$\det \mathbf{A} = 3 \times 1 - 2 \times 2 = 3 - 4 = -1$
The area of S is 1 unit^2
The area of S' is $|-1| \times 1 = 1$ unit^2

31) b)
As $\det \mathbf{A} = -1 < 0$, the orientation of the vertices has been reversed

32) a)
The area of the square $ABCD$ is 1, and the area of the rectangle $A'B'C'D'$ is 2. The orientation of the vertices has been reversed. So $\det \mathbf{A} = -2$

32) b)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 1 & 2 & 2 \\ 2 & 3 & 3 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 6 & 6 & 4 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+2b & a+3b & 2a+3b & 2a+2b \\ c+2d & c+3d & 2c+3d & 2c+2d \end{pmatrix} = \begin{pmatrix} 4 & 6 & 6 & 4 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

$a + 2b = 4$ (1)

$a + 3b = 6$ (2)

$\therefore a = 0, b = 2$

$c + 2d = 1$ (3)

$c + 3d = 1$ (4)

$\therefore c = 1, d = 0$

$\therefore \mathbf{A} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

33)

$\det \mathbf{A} = \frac{3}{5} \times \left(-\frac{3}{5}\right) - \frac{3\sqrt{3}}{5} \times \frac{3\sqrt{3}}{5} = -\frac{9}{25} - \frac{27}{25} = -\frac{36}{25}$

The scale factor of the enlargement is $\sqrt{\frac{36}{25}} = \frac{6}{5}$

34) a)

$\det \mathbf{A} = 4\sqrt{3} \times \frac{\sqrt{3}}{2} - (-4) \times \frac{1}{2} = 6 + 2 = 8$

The scale factor of the stretch is 8

34) b)

$\begin{pmatrix}8 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}a & c\\ b & d\end{pmatrix}=\begin{pmatrix}4\sqrt{3} & -4\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}\Rightarrow\begin{pmatrix}8a & 8b\\ c & d\end{pmatrix}=\begin{pmatrix}4\sqrt{3} & -4\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$

$8a=4\sqrt{3}\Rightarrow a=\frac{\sqrt{3}}{2}$

$8b=-4\Rightarrow b=-\frac{1}{2}$

$c=\frac{1}{2}$ and $d=\frac{\sqrt{3}}{2}$

$\therefore\begin{pmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$ is the rotation matrix

$\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}=\begin{pmatrix}\frac{\sqrt{3}}{2} & -\frac{1}{2}\\ \frac{1}{2} & \frac{\sqrt{3}}{2}\end{pmatrix}$

$\cos\theta=\frac{\sqrt{3}}{2}\Rightarrow\theta=\{30°,330°\}$

$\sin\theta=\frac{1}{2}\Rightarrow\theta=\{30°,150°\}$

So the matrix represents a Rotation, anticlockwise about the origin, by $30°$

35) a)

$\det\mathbf{A}=(k-2)(k+2)-(k-3)(k+3)=(k^2-4)-(k^2-9)=5>0$

So $\det\mathbf{A}>0$ for all values of k

35) b)

As $\det\mathbf{A}>0$, the matrix $\mathbf{A}$ must preserve orientation

36)

$\mathbf{AB}=\begin{pmatrix}8 & -5\\ k & -2\end{pmatrix}\begin{pmatrix}9 & 2\\ 13 & k\end{pmatrix}=\begin{pmatrix}7 & 16-5k\\ 9k-26 & 0\end{pmatrix}$

$\det\mathbf{AB}=-1$ as AB preserves area (so $|\det\mathbf{AB}|=1$) but reverses orientation (must be negative)

$\therefore 7\times 0-(16-5k)(9k-26)=-1\Rightarrow-(-45k^2+274k-416)=-1$

$\Rightarrow-45k^2+274k-416=1\Rightarrow 0=45k^2-274k+417$

$\Rightarrow(45k-139)(k-3)=0\Rightarrow k=\frac{139}{45}$ or $k=3$

37) a)

$\det\mathbf{A}=2\times 6-(-4)\times 3=12+12=24$

37) b)
$\det \mathbf{B} = -1 \times 4 - 2 \times (-8) = -4 + 16 = 12$

37) c)
$\mathbf{AB} = \begin{pmatrix} 2 & -4 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} -1 & 2 \\ -8 & 4 \end{pmatrix} = \begin{pmatrix} 30 & -12 \\ -51 & 30 \end{pmatrix}$

37) d)
$\det \mathbf{AB} = 30 \times 30 - (-12) \times (-51) = 900 - 612 = 288$

37) e)
$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B}$ as $288 = 24 \times 12$

38) a)
$\det \mathbf{AB} = 5 \times (-6) = -30$

38) b)
$\det \mathbf{BA} = (-6) \times 5 = -30$

38) c)
$\det \mathbf{ABA} = 5 \times (-6) \times 5 = -150$

38) d)
$\det \mathbf{BABABA} = (-6) \times 5 \times (-6) \times 5 \times (-6) \times 5 = -27000$

39)
$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} \Rightarrow 20 = \det \mathbf{A} \times 5 \Rightarrow \det \mathbf{A} = 4$
$\det \mathbf{A}^5 = (\det \mathbf{A})^5 = 4^5 = 1024$

40)
$\det \mathbf{AB} = \det \mathbf{I} = 1 \Rightarrow \det \mathbf{A} \times \det \mathbf{B} = 1 \Rightarrow \det \mathbf{A} = \frac{1}{\det \mathbf{B}}$
$\det \mathbf{A}$ and $\det \mathbf{B}$ must be reciprocals of one another.

41)
$\det \mathbf{A} = 4 \times 2 - k \times 6 = 8 - 6k$
$|\mathbf{A}^2| = \det(\mathbf{A}^2) = (\det \mathbf{A})^2 = 16 \Rightarrow (8 - 6k)^2 = 16$
$\Rightarrow 8 - 6k = \pm 4 \Rightarrow k = \frac{8 \pm 4}{6} \Rightarrow k = 2$ or $k = \frac{2}{3}$

42)

$\det \mathbf{A} = 5 \times k - 2 \times (-4) = 5k + 8$

$|\mathbf{A}^3| = \det(\mathbf{A}^3) = (\det \mathbf{A})^3 = 125 \Rightarrow (5k + 8)^3 = 125$

$\Rightarrow 5k + 8 = 5 \Rightarrow k = -\frac{3}{5}$

43)

$\det \mathbf{A} = 4 \times \mathrm{i} - (3 + \mathrm{i}) \times 2 = 4\mathrm{i} - (6 + 2\mathrm{i}) = -6 + 2\mathrm{i}$

$\det \mathbf{B} = (k + 1) \times \mathrm{i} - 3 \times (2 - \mathrm{i}) = (k\mathrm{i} + \mathrm{i}) - (6 - 3\mathrm{i}) = -6 + k\mathrm{i} + 4\mathrm{i} = -6 + (k + 4)\mathrm{i}$

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = (-6 + 2\mathrm{i})(-6 + (k + 4)\mathrm{i}) = 36 - 6(k + 4)\mathrm{i} - 12\mathrm{i} - 2(k + 4)$

$= 36 - 6k\mathrm{i} - 24\mathrm{i} - 12\mathrm{i} - 2k - 8 = 28 - 2k - (6k + 36)\mathrm{i}$

If $\det \mathbf{AB}$ has argument $\frac{\pi}{4}$ then $\mathrm{Re}(\det \mathbf{AB}) = \mathrm{Im}(\det \mathbf{AB})$ and both the real part and imaginary part must be positive.

$\therefore 28 - 2k = -6k - 36 \Rightarrow 4k = -64 \Rightarrow k = -16$

Check: when $k = -16$, $\det \mathbf{AB} = 60 + 60\mathrm{i}$ which has argument $\frac{\pi}{4}$

44)

$\det \mathbf{A} = (2x - 3)(x - 2) - (x - 5)(x + 6) = (2x^2 - 7x + 6) - (x^2 + x - 30)$

$= x^2 - 8x + 36 = (x - 4)^2 - 16 + 36 = (x - 4)^2 + 20$

So the minimum value of $\det \mathbf{A}$ is 20

So the minimum area of S' is $30 \times 20 = 600$ cm^2

Alternative Method:

Let $y = x^2 - 8x + 36$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 8$

Minimum area occurs when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore 2x - 8 = 0 \Rightarrow x = 4$

$y|_{x=4} = 20$

So the minimum value of $\det \mathbf{A}$ is 20

So the minimum area of S' is $30 \times 20 = 600$ cm^2

45)

$\det \mathbf{A} = (2x^2+1)(x^2+3) - (4x-3)(4x+3)$
$= (2x^4+7x^2+3) - (16x^2-9) = 2x^4 - 9x^2 + 12$
Let $y = 2x^4 - 9x^2 + 12$, then $\frac{\mathrm{d}y}{\mathrm{d}x} = 8x^3 - 18x$
Minimum area occurs when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$
$\therefore 8x^3 - 18x = 0 \Rightarrow 4x^3 - 9x = 0 \Rightarrow (4x^2-9)x = 0 \Rightarrow (2x-3)(2x+3)x = 0$
$\Rightarrow x = \frac{3}{2}$ or $x = -\frac{3}{2}$ or $x = 0$
$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x^2 - 18$
$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=-\frac{3}{2}} = 36 > 0$ so $x = -\frac{3}{2}$ is a local minimum
$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0} = -18 < 0$ so $x = 0$ is a local maximum
$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\frac{3}{2}} = 36 > 0$ so $x = \frac{3}{2}$ is a local minimum
$y|_{x=-\frac{3}{2}} = \frac{15}{8}$ and $y|_{x=\frac{3}{2}} = \frac{15}{8}$
So the minimum value of $\det \mathbf{A}$ is $\frac{15}{8}$
So the minimum area of S' is $16 \times \frac{15}{8} = 30$ cm^2

46)

$\det \mathbf{A} = (k+1)(k+2) - (k-3)(k-4) = (k^2+3k+2) - (k^2-7k+12)$
$= 10k - 10$
$\det \mathbf{B} = (2k-1)(2k-1) - (3k+1)(1-3k) = (4k^2-4k+1) - (-9k^2+1)$
$= 13k^2 - 4k$
$\det \mathbf{AB} = 0 \Rightarrow \det \mathbf{A} \times \det \mathbf{B} = 0 \Rightarrow (10k-10)(13k^2-4k) = 0$
$\Rightarrow (k-1)(13k-4)k = 0 \Rightarrow k = 0$ or $k = 1$ or $k = \frac{4}{13}$

47)

$\det \mathbf{A} = (2k+3)(k-4) - (k+1)(k-2) = (2k^2-5k-12) - (k^2-k-2)$
$= k^2 - 4k - 10$
$\det \mathbf{B} = (3k+1)(k+2) - (k-3)(k+4) = (3k^2+7k+2) - (k^2+k-12)$
$= 2k^2 + 6k + 14$
$\det \mathbf{A} = \det \mathbf{B} \Rightarrow k^2 - 4k - 10 = 2k^2 + 6k + 14 \Rightarrow 0 = k^2 + 10k + 24$
$\Rightarrow 0 = (k+6)(k+4) \Rightarrow k = -6$ or $k = -4$

48) a) i)

$$\begin{vmatrix} ka & b \\ kc & d \end{vmatrix} = ka \times d - b \times kc = kad - kbc = k(ad-bc) = k\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

48) a) ii)

$$\begin{vmatrix} a & kb \\ c & kd \end{vmatrix} = a \times kd - kb \times c = kad - kbc = k(ad - bc) = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

48) a) iii)

$$\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = ka \times d - kb \times c = kad - kbc = k(ad - bc) = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

48) a) iv)

$$\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = ka \times kd - kb \times kc = k^2ad - k^2bc = k^2(ad - bc) = k^2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

48) b) i)

$$\begin{vmatrix} 6 & 8 \\ -2 & 5 \end{vmatrix} = \begin{vmatrix} 2 \times 3 & 2 \times 4 \\ -2 & 5 \end{vmatrix} = 2 \begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 2 \times 23 = 46$$

48) b) ii)

$$\begin{vmatrix} 3 & 12 \\ -2 & 15 \end{vmatrix} = \begin{vmatrix} 3 & 3 \times 4 \\ -2 & 3 \times 5 \end{vmatrix} = 3 \begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 3 \times 23 = 69$$

48) b) iii)

$$\begin{vmatrix} 30 & 4 \\ -20 & 5 \end{vmatrix} = \begin{vmatrix} 10 \times 3 & 4 \\ 10 \times (-2) & 5 \end{vmatrix} = 10 \begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 10 \times 23 = 230$$

48) b) iv)

$$\begin{vmatrix} 15 & 20 \\ -20 & 50 \end{vmatrix} = \begin{vmatrix} 5 \times 3 & 5 \times 4 \\ 10 \times (-2) & 10 \times 5 \end{vmatrix} = 5 \times \begin{vmatrix} 3 & 4 \\ 10 \times (-2) & 10 \times 5 \end{vmatrix} = 5 \times 10 \times \begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix}$$
$$= 5 \times 10 \times 23 = 1150$$

49) a)

Calculator:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 12$$

By hand:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 1 \times \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 0 & 6 \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$
$$= 1 \times (2 \times 6 - 0 \times 0) - 0 + 0 = 1 \times 12 = 12$$

49) b)

Calculator:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 36$$

By hand:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 3 \times \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 0 & 6 \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$

$$= 3 \times (2 \times 6 - 0 \times 0) - 0 + 0 = 3 \times 12 = 36$$

49) c)

Calculator:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 27$$

By hand:

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 3 \times \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 3 \\ 0 & 6 \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix}$$

$$= 3 \times (2 \times 6 - 3 \times 1) - 0 + 0 = 3 \times 9 = 27$$

49) d)

Calculator:

$$\begin{vmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 27$$

By hand:

$$\begin{vmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{vmatrix} = 3 \times \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 3 \\ 0 & 6 \end{vmatrix} + 1 \times \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix}$$

$$= 3 \times (2 \times 6 - 3 \times 1) - 0 + 1 \times (0 \times 1 - 2 \times 0) = 3 \times 9 + 1 \times 0 = 27$$

49) e)

Calculator:

$$\begin{vmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{vmatrix} = 9$$

By hand:

$$\begin{vmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{vmatrix} = 3 \times \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 3 \\ 1 & 6 \end{vmatrix} + 1 \times \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= 3 \times (2 \times 6 - 3 \times 1) - 2 \times (2 \times 6 - 3 \times 1) + 1 \times (2 \times 1 - 2 \times 1)$$

$$= 3 \times 9 - 2 \times 9 + 1 \times 0 = 9$$

50)

Calculator:

$$\begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} = -4$$

By hand:

$$\begin{vmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{vmatrix} = 1 \times \begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} - 1 \times \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} + 2 \times \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= 1 \times (1 \times 4 - 1 \times 1) - 1 \times (3 \times 4 - 1 \times 1) + 2 \times (3 \times 1 - 1 \times 1)$$

$$= 1 \times 3 - 1 \times 11 + 2 \times 2 = -4$$

Volume of $C' = 4 \times 33 = 132$ cm^3

51) a)

Volume of C is $5 \times 6 \times 8 = 240$ units^3

51) b)

Calculator:

$$\begin{vmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{vmatrix} = \frac{1}{2}$$

By hand:

$$\begin{vmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{vmatrix} = 2 \times \begin{vmatrix} 3 & 0 \\ 0 & \frac{1}{12} \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 0 & \frac{1}{12} \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 3 \\ 0 & 0 \end{vmatrix}$$

$$= 2 \times \left(3 \times \frac{1}{12} - 0 \times 0\right) - 0 + 0 = 2 \times \frac{1}{4} = \frac{1}{2}$$

51) c)

Volume of C' is $240 \times \frac{1}{2} = 120$ units^3

52)

Calculator:

$$\begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{vmatrix} = 50$$

By hand:

$$\begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{vmatrix} = 2 \times \begin{vmatrix} 0 & 6 \\ 2 & 4 \end{vmatrix} - (-3) \times \begin{vmatrix} 1 & 6 \\ -3 & 4 \end{vmatrix} + 4 \times \begin{vmatrix} 1 & 0 \\ -3 & 2 \end{vmatrix}$$

$$= 2 \times (0 \times 4 - 6 \times 2) - (-3) \times \big(1 \times 4 - 6 \times (-3)\big) + 4 \times \big(1 \times 2 - 0 \times (-3)\big)$$

$$= 2 \times (-12) + 3 \times 22 + 4 \times 2 = 50$$

Volume of S' is $9 \times 50 = 450$ cm^3

53) a)

Calculator:

$$\begin{vmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{vmatrix} = -1$$

By hand:

$$\begin{vmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{vmatrix} = 3 \times \begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} - 2 \times \begin{vmatrix} 1 & -1 \\ 3 & -1 \end{vmatrix} + 0 \times \begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix}$$

$$= 3 \times (2 \times (-1) - (-1) \times 3) - 2 \times (1 \times (-1) - (-1) \times 3) + 0$$

$$= 3 \times 1 - 2 \times 2 = -1$$

53) b)

As $\det \mathbf{A} = -1$, the transformation preserves volume but reverses orientation.

54) a)

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & \frac{1}{9} \end{vmatrix} = 3\left(6 \times \tfrac{1}{9} - 0 \times 0\right) - 0\left(0 \times \tfrac{1}{9} - 0 \times 0\right) + 0(0 \times 0 - 6 \times 0)$$

$$= 3 \times 6 \times \tfrac{1}{9} = 2$$

So the volume scale factor is 2

54) b)

$$\begin{vmatrix} -2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 3 \end{vmatrix} = -2(18 \times 3 - 0 \times 0) - 0(0 \times 3 - 0 \times 0) + 0(0 \times 0 - 18 \times 0)$$

$= -2 \times 18 \times 3 = -108$

So the volume scale factor is 108

54) c)

$$\begin{vmatrix} 5 & 1 & 7 \\ 2 & 0 & 0 \\ 5 & 0 & 1 \end{vmatrix} = 5(0 \times 1 - 0 \times 0) - 1(2 \times 1 - 0 \times 5) + 7(2 \times 0 - 0 \times 5)$$

$= 0 - 2 + 0 = -2$

So the volume scale factor is 2

54) d)

$$\begin{vmatrix} 3 & -2 & -4 \\ 2 & 4 & -3 \\ 3 & 1 & 3 \end{vmatrix} = 3(4 \times 3 - (-3) \times 1) + 2(2 \times 3 - (-3) \times 3) - 4(2 \times 1 - 4 \times 3)$$

$= 45 + 30 + 40 = 115$

So the volume scale factor is 115

55)

$$\begin{vmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{vmatrix} = \frac{1}{3}\left(\frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{2}} - 0 \times 0\right) - 0\left(0 \times \frac{1}{\sqrt{2}} - 0 \times 0\right) + 0\left(0 \times 0 - \frac{1}{\sqrt{3}} \times 0\right)$$

$= \frac{1}{3\sqrt{6}} = \frac{\sqrt{6}}{18}$

Radius of S is $\sqrt{27} = 3\sqrt{3}$ units

Volume of $S = \frac{4}{3}\pi \times \left(3\sqrt{3}\right)^3 = 108\sqrt{3}\pi$ units^3

Volume of $S' = \frac{\sqrt{6}}{18} \times 108\sqrt{3}\pi = 18\sqrt{2}\pi$ units^3

56)

Calculator:

$\det \mathbf{A} = 13$, so $(\det \mathbf{A})^2 = 169$

$$\mathbf{A}^2 = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}\begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{pmatrix}$$

$\det(\mathbf{A}^2) = 169$

$\therefore \det(\mathbf{A}^2) = (\det \mathbf{A})^2$

By hand:

$$\det \mathbf{A} = \begin{vmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{vmatrix} = 2 \times \begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix} - 3 \times \begin{vmatrix} -4 & 1 \\ -3 & 1 \end{vmatrix} + 8 \times \begin{vmatrix} -4 & 3 \\ -3 & 2 \end{vmatrix}$$

$$= 2 \times (3 \times 1 - 1 \times 2) - 3 \times \big((-4) \times 1 - 1 \times (-3)\big) + 8 \times \big((-4) \times 2 - 3 \times (-3)\big)$$

$= 2 \times 1 - 3 \times (-1) + 8 \times 1 = 13$, so $(\det \mathbf{A})^2 = 169$

$$\mathbf{A}^2 = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}\begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{pmatrix}$$

$$\det(\mathbf{A}^2) = \begin{vmatrix} -32 & 31 & 27 \\ -23 & -1 & -28 \\ -17 & -1 & -21 \end{vmatrix}$$

$$= -32 \times \begin{vmatrix} -1 & -28 \\ -1 & -21 \end{vmatrix} - 31 \times \begin{vmatrix} -23 & -28 \\ -17 & -21 \end{vmatrix} + 27 \times \begin{vmatrix} -23 & -1 \\ -17 & -1 \end{vmatrix}$$

$$= -32 \times \big((-1) \times (-21) - (-28) \times (-1)\big)$$

$$-31 \times \big((-23) \times (-21) - (-28) \times (-17)\big)$$

$$+27 \times \big((-23) \times (-1) - (-1) \times (-17)\big)$$

$= -32 \times (-7) - 31 \times 7 + 27 \times 6 = 169$

$\therefore \det(\mathbf{A}^2) = (\det \mathbf{A})^2$

57)

Calculator:

$\det \mathbf{A} = -1$, so $(\det \mathbf{A})^3 = -1$

$$\mathbf{A}^3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{pmatrix}$$

$\det(\mathbf{A}^3) = -1$

$\therefore \det(\mathbf{A}^3) = (\det \mathbf{A})^3$

By hand:

$$\det \mathbf{A} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 1 \times \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + 1 \times \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix}$$

$= 1 \times (1 \times 1 - 1 \times 1) - 0 + 1 \times (0 \times 1 - 1 \times 1)$

$= 1 \times 0 + 1 \times (-1) = -1$, so $(\det \mathbf{A})^3 = -1$

$$\mathbf{A}^3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{pmatrix}$$

$$\det(\mathbf{A}^3) = \begin{vmatrix} 4 & 3 & 5 \\ 3 & 4 & 5 \\ 5 & 5 & 7 \end{vmatrix} = 4 \times \begin{vmatrix} 4 & 5 \\ 5 & 7 \end{vmatrix} - 3 \times \begin{vmatrix} 3 & 5 \\ 5 & 7 \end{vmatrix} + 5 \times \begin{vmatrix} 3 & 4 \\ 5 & 5 \end{vmatrix}$$

$= 4 \times (4 \times 7 - 5 \times 5) - 3 \times (3 \times 7 - 5 \times 5) + 5 \times (3 \times 5 - 4 \times 5)$

$= 4 \times 3 - 3 \times (-4) + 5 \times (-5) = -1$

$\therefore \det(\mathbf{A}^3) = (\det \mathbf{A})^3$

58) a)

Calculator:

$$\det \mathbf{A} = \begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & -5 \\ 0 & 4 & 0 \end{vmatrix} = 0$$

By hand:

$$\det \mathbf{A} = \begin{vmatrix} 0 & 2 & 0 \\ 3 & 0 & -5 \\ 0 & 4 & 0 \end{vmatrix} = 0 \times \begin{vmatrix} 0 & -5 \\ 4 & 0 \end{vmatrix} - 2 \times \begin{vmatrix} 3 & -5 \\ 0 & 0 \end{vmatrix} + 0 \times \begin{vmatrix} 3 & 0 \\ 0 & 4 \end{vmatrix}$$

$= 0 - 2 \times (3 \times 0 - (-5) \times 0) + 0 = -2 \times 0 = 0$

$\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3

58) b)

Calculator:

$$\det \mathbf{B} = \begin{vmatrix} 5 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{vmatrix} = 34$$

By hand:

$$\det \mathbf{B} = \begin{vmatrix} 5 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{vmatrix} = 5 \times \begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 3 & 4 \end{vmatrix} + 1 \times \begin{vmatrix} 0 & 2 \\ 3 & 0 \end{vmatrix}$$

$= 5 \times (2 \times 4 - 0 \times 0) - 0 + 1 \times (0 \times 0 - 2 \times 3) = 5 \times 8 + 1 \times (-6) = 34$

$\therefore$ the volume of S' is $12 \times 34 = 408$ cm^3

58) c)

Calculator:

$$\det \mathbf{C} = \begin{vmatrix} -1 & 2 & 1 \\ 3 & -2 & 4 \\ -9 & 1 & 4 \end{vmatrix} = -99$$

By hand:

$$\det \mathbf{C} = \begin{vmatrix} -1 & 2 & 1 \\ 3 & -2 & 4 \\ -9 & 1 & 4 \end{vmatrix} = -1 \times \begin{vmatrix} -2 & 4 \\ 1 & 4 \end{vmatrix} - 2 \times \begin{vmatrix} 3 & 4 \\ -9 & 4 \end{vmatrix} + 1 \times \begin{vmatrix} 3 & -2 \\ -9 & 1 \end{vmatrix}$$

$= -1 \times \big((-2) \times 4 - 4 \times 1\big) - 2 \times \big(3 \times 4 - 4 \times (-9)\big) + 1 \times \big(3 \times 1 - (-2) \times (-9)\big)$

$= -1 \times (-12) - 2 \times 48 + 1 \times (-15) = -99$

$\therefore$ the volume of S' is $12 \times 99 = 1188$ cm^3

58) d)

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = 0 \times 34 = 0$

$\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3

58) e)

$\det \mathbf{AC} = \det \mathbf{A} \times \det \mathbf{C} = 0 \times -99 = 0$

$\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3

58) f)

$\det \mathbf{BC} = \det \mathbf{B} \times \det \mathbf{C} = 34 \times -99 = -3366$

$\therefore$ the volume of S' is $12 \times 3366 = 40392$ cm^3

58) g)

$\det \mathbf{ABC} = \det \mathbf{A} \times \det \mathbf{B} \times \det \mathbf{C} = 0 \times 34 \times -99 = 0$

$\therefore$ the volume of S' is $12 \times 0 = 0$ cm^3

59) a)

$$\begin{vmatrix} k & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & k \end{vmatrix} = k \times \begin{vmatrix} 2 & 0 \\ 0 & k \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 0 & k \end{vmatrix} + 0 \times \begin{vmatrix} 0 & 2 \\ 0 & 0 \end{vmatrix}$$

$$= k \times (2 \times k - 0 \times 0) - 0 + 0 = k \times 2k = 2k^2$$

59) b)

$$\begin{vmatrix} k & 0 & 1 \\ 0 & k & 0 \\ 4 & 0 & k \end{vmatrix} = k \times \begin{vmatrix} k & 0 \\ 0 & k \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 4 & k \end{vmatrix} + 1 \times \begin{vmatrix} 0 & k \\ 4 & 0 \end{vmatrix}$$

$$= k \times (k \times k - 0 \times 0) - 0 + 1 \times (0 \times 0 - k \times 4) = k \times k^2 + 1 \times (-4k) = k^3 - 4k$$

59) c)

$$\begin{vmatrix} k & 1 & 0 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{vmatrix} = k \times \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} - 1 \times \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} + 0 \times \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= k \times (2 \times 4 - 3 \times 1) - 1 \times (1 \times 4 - 3 \times 2) + 0$$

$$= k \times 5 - 1 \times (-2) + 0 = 5k + 2$$

59) d)

$$\begin{vmatrix} k & 0 & k \\ 3 & 5 & -1 \\ 2 & -3 & 2 \end{vmatrix} = k \times \begin{vmatrix} 5 & -1 \\ -3 & 2 \end{vmatrix} - 0 \times \begin{vmatrix} 3 & -1 \\ 2 & 2 \end{vmatrix} + k \times \begin{vmatrix} 3 & 5 \\ 2 & -3 \end{vmatrix}$$

$$= k \times \big(5 \times 2 - (-1) \times (-3)\big) - 0 + k \times (3 \times (-3) - 5 \times 2)$$

$$= k \times 7 + k \times (-19) = -12k$$

59) e)

$$\begin{vmatrix} 1 & 2 & 4 \\ k & 2 & 1 \\ -4 & 3 & 5 \end{vmatrix} = 1 \times \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix} - 2 \times \begin{vmatrix} k & 1 \\ -4 & 5 \end{vmatrix} + 4 \times \begin{vmatrix} k & 2 \\ -4 & 3 \end{vmatrix}$$

$$= 1 \times (2 \times 5 - 1 \times 3) - 2 \times \big(k \times 5 - 1 \times (-4)\big) + 4 \times \big(k \times 3 - 2 \times (-4)\big)$$

$$= 1 \times 7 - 2 \times (5k + 4) + 4 \times (3k + 8) = 2k + 31$$

59) f)

$$\begin{vmatrix} 4 & k & k \\ k & 3 & 2 \\ k & 2 & 3 \end{vmatrix} = 4 \times \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} - k \times \begin{vmatrix} k & 2 \\ k & 3 \end{vmatrix} + k \times \begin{vmatrix} k & 3 \\ k & 2 \end{vmatrix}$$

$$= 4 \times (3 \times 3 - 2 \times 2) - k \times (k \times 3 - 2 \times k) + k \times (k \times 2 - 3 \times k)$$

$$= 4 \times 5 - k \times k + k \times (-k) = 20 - 2k^2$$

59) g)

$$\begin{vmatrix} 5 & 3k & 3 \\ -2 & 1 & 4k \\ 2k & 3 & -5 \end{vmatrix} = 5 \times \begin{vmatrix} 1 & 4k \\ 3 & -5 \end{vmatrix} - 3k \times \begin{vmatrix} -2 & 4k \\ 2k & -5 \end{vmatrix} + 3 \times \begin{vmatrix} -2 & 1 \\ 2k & 3 \end{vmatrix}$$

$$= 5 \times (1 \times (-5) - 4k \times 3) - 3k \times (-2 \times (-5) - 4k \times 2k) + 3 \times (-2 \times 3 - 1 \times 2k)$$

$$= 5 \times (-5 - 12k) - 3k \times (10 - 8k^2) + 3 \times (-6 - 2k)$$

$$= -25 - 60k - 30k + 24k^3 - 18 - 6k = 24k^3 - 96k - 43$$

60) a)

$$\begin{vmatrix} k & 1 & 0 \\ 2 & 4 & 3 \\ -1 & 2 & 3 \end{vmatrix} = 1 \Rightarrow k \times \begin{vmatrix} 4 & 3 \\ 2 & 3 \end{vmatrix} - 1 \times \begin{vmatrix} 2 & 3 \\ -1 & 3 \end{vmatrix} + 0 \times \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = 1$$

$$\Rightarrow k \times (4 \times 3 - 3 \times 2) - 1 \times \big(2 \times 3 - 3 \times (-1)\big) + 0 = 1$$

$$\Rightarrow k \times 6 - 1 \times 9 = 1 \Rightarrow 6k - 9 = 1 \Rightarrow k = \frac{5}{3}$$

60) b)

$$\begin{vmatrix} 3 & k & 4 \\ -1 & 2 & -5 \\ 3 & 2 & 7 \end{vmatrix} = 8 \Rightarrow 3 \times \begin{vmatrix} 2 & -5 \\ 2 & 7 \end{vmatrix} - k \times \begin{vmatrix} -1 & -5 \\ 3 & 7 \end{vmatrix} + 4 \times \begin{vmatrix} -1 & 2 \\ 3 & 2 \end{vmatrix} = 8$$

$$\Rightarrow 3 \times (2 \times 7 - (-5) \times 2) - k \times \big((-1) \times 7 - (-5) \times 3\big) + 4 \times \big((-1) \times 2 - 2 \times 3\big) = 8$$

$$\Rightarrow 3 \times 24 - k \times 8 + 4 \times (-8) = 8 \Rightarrow 40 - 8k = 8 \Rightarrow k = 4$$

60) c)

$$\begin{vmatrix} 2 & 3 & -5 \\ -2 & k & 2 \\ 1 & -4 & 1 \end{vmatrix} = -33 \Rightarrow 2 \times \begin{vmatrix} k & 2 \\ -4 & 1 \end{vmatrix} - 3 \times \begin{vmatrix} -2 & 2 \\ 1 & 1 \end{vmatrix} + (-5) \times \begin{vmatrix} -2 & k \\ 1 & -4 \end{vmatrix} = -33$$

$$\Rightarrow 2 \times \big(k \times 1 - 2 \times (-4)\big) - 3 \times (-2 \times 1 - 2 \times 1) - 5 \times (-2 \times (-4) - k \times 1) = -33$$

$$\Rightarrow 2 \times (k + 8) - 3 \times (-4) - 5 \times (8 - k) = -33 \Rightarrow 2k + 16 + 12 - 40 + 5k = -33$$

$$\Rightarrow 7k - 12 = -33 \Rightarrow k = -3$$

60) d)

$$\begin{vmatrix} k & 2 & 5 \\ 9 & 8 & 2 \\ k & 3 & 1 \end{vmatrix} = 49 \Rightarrow k \times \begin{vmatrix} 8 & 2 \\ 3 & 1 \end{vmatrix} - 2 \times \begin{vmatrix} 9 & 2 \\ k & 1 \end{vmatrix} + 5 \times \begin{vmatrix} 9 & 8 \\ k & 3 \end{vmatrix} = 49$$

$$\Rightarrow k \times (8 \times 1 - 2 \times 3) - 2 \times (9 \times 1 - 2 \times k) + 5 \times (9 \times 3 - 8 \times k) = 49$$

$$\Rightarrow k \times 2 - 2 \times (9 - 2k) + 5 \times (27 - 8k) = 49 \Rightarrow 2k - 18 + 4k + 135 - 40k = 49$$

$$\Rightarrow 117 - 34k = 49 \Rightarrow k = 2$$

60) e)

$$\begin{vmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k+1 \end{vmatrix} = 150 \Rightarrow k \times \begin{vmatrix} k & 0 \\ 0 & k+1 \end{vmatrix} - 0 \times \begin{vmatrix} 0 & 0 \\ 0 & k+1 \end{vmatrix} + 0 \times \begin{vmatrix} 0 & k \\ 0 & 0 \end{vmatrix} = 150$$

$$\Rightarrow k \times (k \times (k+1) - 0 \times 0) - 0 + 0 = 150$$

$$\Rightarrow k^2(k+1) = 150 \Rightarrow k^3 + k^2 - 150 = 0 \Rightarrow k = 5$$

60) f)

$$\begin{vmatrix} 2k & 1 & 3k \\ 1 & 5 & 1 \\ 3k & 1 & 2k \end{vmatrix} = -23 \Rightarrow 2k \times \begin{vmatrix} 5 & 1 \\ 1 & 2k \end{vmatrix} - 1 \times \begin{vmatrix} 1 & 1 \\ 3k & 2k \end{vmatrix} + 3k \times \begin{vmatrix} 1 & 5 \\ 3k & 1 \end{vmatrix} = -23$$

$$\Rightarrow 2k \times (5 \times 2k - 1 \times 1) - 1 \times (1 \times 2k - 1 \times 3k) + 3k \times (1 \times 1 - 5 \times 3k) = -23$$

$$\Rightarrow 2k \times (10k - 1) - 1 \times (-k) + 3k \times (1 - 15k) = -23$$

$$\Rightarrow 20k^2 - 2k + k + 3k - 45k^2 = -23$$

$$\Rightarrow 0 = 25k^2 - 2k - 23 = 0 \Rightarrow (25k + 23)(k - 1) = 0 \Rightarrow k = -\frac{23}{25} \text{ or } k = 1$$

60) g)

$$\begin{vmatrix} k+1 & 2 & 2k \\ 1 & k & 3 \\ k & 0 & k \end{vmatrix} = -2 \Rightarrow (k+1) \times \begin{vmatrix} k & 3 \\ 0 & k \end{vmatrix} - 2 \times \begin{vmatrix} 1 & 3 \\ k & k \end{vmatrix} + 2k \times \begin{vmatrix} 1 & k \\ k & 0 \end{vmatrix} = -2$$

$$\Rightarrow (k+1) \times (k \times k - 3 \times 0) - 2 \times (1 \times k - 3 \times k) + 2k \times (1 \times 0 - k \times k) = -2$$

$$\Rightarrow (k+1) \times k^2 - 2 \times (-2k) + 2k \times (-k^2) = -2$$

$$\Rightarrow k^3 + k^2 + 4k - 2k^3 = -2 \Rightarrow k^3 - k^2 - 4k - 2 = 0$$

$$(k+1)(k^2 - 2k - 2) = 0 \Rightarrow k = -1 \text{ or } k = 1 \pm \sqrt{3}$$

61) a)

When $k = 5$, $\mathbf{A} = \begin{pmatrix} 0 & 5 & 2 \\ -23 & 6 & 5 \\ 2 & \frac{1}{2} & 0 \end{pmatrix}$

$$\det \mathbf{A} = 0\left(6 \times 0 - 5 \times \frac{1}{2}\right) - 5(-23 \times 0 - 5 \times 2) + 2\left(-23 \times \frac{1}{2} - 6 \times 2\right)$$

$$= 0 - 5(-10) + 2\left(-\frac{47}{2}\right) = 3$$

$$\therefore p = 3$$

61) b)

$$\det \mathbf{A} = 3 \Rightarrow \begin{vmatrix} 0 & k & 2 \\ -23 & k+1 & k \\ k-3 & \frac{1}{2} & 0 \end{vmatrix} = 3$$

$$\Rightarrow 0\left((k+1)\times 0 - k\times\frac{1}{2}\right) - k\left(-23\times 0 - k\times(k-3)\right) + 2\left(-23\times\frac{1}{2} - (k+1)\times(k-3)\right) = 3$$

$$\Rightarrow 0 - k\left(-k(k-3)\right) + 2\left(-\frac{23}{2} - (k+1)(k-3)\right) = 3$$

$$\Rightarrow k^2(k-3) - 23 - 2(k^2 - 2k - 3) = 3$$

$$\Rightarrow k^3 - 3k^2 - 23 - 2k^2 + 4k + 6 - 3 = 0$$

$$\Rightarrow k^3 - 5k^2 + 4k - 20 = 0$$

$$\Rightarrow (k-5)(k^2+4) = 0$$

$$\Rightarrow k = 5 \text{ or } k = \pm 2\text{i}$$

62)

$$\det \mathbf{A} = \begin{vmatrix} k-2 & 2k & 3k+3 \\ 2k-4 & 4k & k+1 \\ 2-k & 8k & 9k+9 \end{vmatrix}$$

$$= (k-2)\times\begin{vmatrix} 4k & k+1 \\ 8k & 9k+9 \end{vmatrix} - 2k\times\begin{vmatrix} 2k-4 & k+1 \\ 2-k & 9k+9 \end{vmatrix} + (3k+3)\times\begin{vmatrix} 2k-4 & 4k \\ 2-k & 8k \end{vmatrix}$$

$$= (k-2)\times(4k\times(9k+9) - (k+1)\times 8k)$$

$$-2k\times\left((2k-4)\times(9k+9) - (k+1)\times(2-k)\right)$$

$$+(3k+3)\times\left((2k-4)\times 8k - 4k\times(2-k)\right)$$

$$= (k-2)(4k(9k+9) - 8(k+1)k)$$

$$-2k\left((2k-4)(9k+9) - (k+1)(2-k)\right)$$

$$+(3k+3)\left(8(2k-4)k - 4k(2-k)\right)$$

$$= (k-2)(36k(k+1) - 8(k+1)k)$$

$$-2k\left(18(k-2)(k+1) + (k+1)(k-2)\right)$$

$$+3(k+1)\left(16(k-2)k + 4k(k-2)\right)$$

$$= 36(k-2)(k+1)k - 8(k-2)(k+1)k - 36(k-2)(k+1)k - 2(k-2)(k+1)k + 48(k-2)(k+1)k + 12(k-2)(k+1)k$$

$$= 50(k-2)(k+1)k$$

Alternative Method:

$$\begin{vmatrix} k-2 & 2k & 3k+3 \\ 2k-4 & 4k & k+1 \\ 2-k & 8k & 9k+9 \end{vmatrix} = \begin{vmatrix} k-2 & 2k & 3(k+1) \\ 2(k-2) & 4k & k+1 \\ -(k-2) & 8k & 9(k+1) \end{vmatrix} = (k-2)\begin{vmatrix} 1 & 2k & 3(k+1) \\ 2 & 4k & k+1 \\ -1 & 8k & 9(k+1) \end{vmatrix}$$

$$= 2(k-2)k\begin{vmatrix} 1 & 1 & 3(k+1) \\ 2 & 2 & k+1 \\ -1 & 4 & 9(k+1) \end{vmatrix} = 2(k-2)(k+1)k\begin{vmatrix} 1 & 1 & 3 \\ 2 & 2 & 1 \\ -1 & 4 & 9 \end{vmatrix}$$

$$= 2(k-2)(k+1)k \times 25 = 50(k-2)(k+1)k$$

63)
A singular matrix is a square matrix where the determinant is zero (and hence has no inverse)

64) a)
$\begin{vmatrix} 4 & 2 \\ 4 & 2 \end{vmatrix} = 4 \times 2 - 2 \times 4 = 0$
So $\begin{pmatrix} 4 & 2 \\ 4 & 2 \end{pmatrix}$ is singular

64) b)
$\begin{vmatrix} 3 & 18 \\ 1 & 6 \end{vmatrix} = 3 \times 6 - 18 \times 1 = 0$
So $\begin{pmatrix} 3 & 18 \\ 1 & 6 \end{pmatrix}$ is singular

64) c)
$\begin{vmatrix} -2 & -5 \\ 4 & 10 \end{vmatrix} = -2 \times 10 - (-5) \times 4 = 0$
So $\begin{pmatrix} -2 & -5 \\ 4 & 10 \end{pmatrix}$ is singular

64) d)
$\begin{vmatrix} 6 & -6 \\ -8 & -8 \end{vmatrix} = 6 \times (-8) - (-6) \times (-8) = -96$
So $\begin{pmatrix} 6 & -6 \\ -8 & -8 \end{pmatrix}$ is not singular

64) e)
$\begin{vmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{vmatrix} = \sqrt{3} \times 2\sqrt{3} - \left(-\sqrt{6}\right) \times \left(-\sqrt{6}\right) = 0$
So $\begin{pmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{pmatrix}$ is singular

64) f)
$\begin{vmatrix} k+3 & k \\ 2k+6 & 2k \end{vmatrix} = (k+3) \times 2k - k \times (2k+6) = 2k^2 + 6k - 2k^2 - 6k = 0$
So $\begin{pmatrix} k+3 & k \\ 2k+6 & 2k \end{pmatrix}$ is singular

65) a)
$\det \mathbf{A} = 4 \times 6 - 3 \times 8 = 0$
$\therefore$ the area of S' is $16 \times 0 = 0$ cm^2

65) b)

$\begin{pmatrix}4 & 3\\ 8 & 6\end{pmatrix}\begin{pmatrix}a\\ b\end{pmatrix}=\begin{pmatrix}4a+3b\\ 8a+6b\end{pmatrix}$

$y=\frac{8a+6b}{4a+3b}x \Rightarrow y=\frac{2(4a+3b)}{4a+3b}x \Rightarrow y=2x$

66) a)

$\begin{vmatrix}k-3 & 2k+9\\ 8 & -4\end{vmatrix}=0 \Rightarrow (k-3)\times(-4)-(2k+9)\times 8=0$

$\Rightarrow -4k+12-16k-72=0 \Rightarrow -20k-60=0 \Rightarrow k=-3$

66) b)

$\begin{vmatrix}k-1 & k+5\\ k+2 & k+3\end{vmatrix}=0 \Rightarrow (k-1)\times(k+3)-(k+5)\times(k+2)=0$

$\Rightarrow (k^2+2k-3)-(k^2+7k+10)=0 \Rightarrow -5k-13=0 \Rightarrow k=-\frac{13}{5}$

66) c)

$\begin{vmatrix}k+6 & k-4\\ 3-k & 8-k\end{vmatrix}=0 \Rightarrow (k+6)\times(8-k)-(k-4)\times(3-k)=0$

$\Rightarrow (-k^2+2k+48)-(-k^2+7k-12)=0 \Rightarrow -5k+60=0 \Rightarrow k=12$

66) d)

$\begin{vmatrix}2k+3 & k+1\\ k+2 & k+2\end{vmatrix}=0 \Rightarrow (2k+3)\times(k+2)-(k+1)\times(k+2)=0$

$\Rightarrow \big((2k+3)-(k+1)\big)(k+2)=0 \Rightarrow (k+2)(k+2)=0$

$\Rightarrow (k+2)^2=0 \Rightarrow k=-2$

66) e)

$\begin{vmatrix}3k-2 & k+4\\ k-2 & k+3\end{vmatrix}=0 \Rightarrow (3k-2)\times(k+3)-(k+4)\times(k-2)=0$

$\Rightarrow (3k^2+7k-6)-(k^2+2k-8)=0 \Rightarrow 2k^2+5k+2=0$

$\Rightarrow (2k+1)(k+2)=0 \Rightarrow k=-\frac{1}{2}$ or $k=-2$

67)

$\det\mathbf{A}=3\times 4-p\times(-2)=2p+12$

$\det\mathbf{B}=4\times q-2\times 6=4q-12$

$\det\mathbf{AB}=0 \Rightarrow \det\mathbf{A}\times\det\mathbf{B}=0 \Rightarrow (2p+12)(4q-12)=0$

$\Rightarrow 8pq-24p+48q-144=0 \Rightarrow 8pq+48q=24p+144 \Rightarrow pq+6q=3p+18$

68) a)

$\det \mathbf{M} = 4 \times \frac{5}{2} - 30 \times \frac{1}{3} = 10 - 10 = 0$

$\therefore$ the matrix $\mathbf{M}$ is singular

68) b)

$\det \mathbf{N} = 10 \times \left(-\frac{1}{3}\right) - \frac{2}{3} \times (-155) = -\frac{10}{3} + \frac{310}{3} = 100 \neq 0$

$\therefore$ the matrix $\mathbf{N}$ is not singular

68) c)

$\det \mathbf{MN} = \det \mathbf{M} \times \det \mathbf{N} = 0 \times 100 = 0$

$\therefore$ the matrix $\mathbf{MN}$ is singular

68) d)

$\det \mathbf{NM} = \det \mathbf{N} \times \det \mathbf{M} = 100 \times 0 = 0$

$\therefore$ the matrix $\mathbf{NM}$ is singular

69) a)

$$\begin{vmatrix} 6 & 3 & 4 \\ 3 & 0 & 1 \\ 2 & -1 & 0 \end{vmatrix} = 6 \times \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} - 3 \times \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix} + 4 \times \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix}$$

$= 6 \times \big(0 \times 0 - 1 \times (-1)\big) - 3 \times (3 \times 0 - 1 \times 2) + 4 \times (3 \times (-1) - 0 \times 2)$

$= 6 \times 1 - 3 \times (-2) + 4 \times (-3) = 6 + 6 - 12 = 0$

So the matrix is singular

69) b)

$$\begin{vmatrix} 0 & -6 & 4 \\ -1 & 1 & 0 \\ 0 & 3 & -2 \end{vmatrix} = 0 \times \begin{vmatrix} 1 & 0 \\ 3 & -2 \end{vmatrix} - (-6) \times \begin{vmatrix} -1 & 0 \\ 0 & -2 \end{vmatrix} + 4 \times \begin{vmatrix} -1 & 1 \\ 0 & 3 \end{vmatrix}$$

$= 0 \times (1 \times (-2) - 0 \times 3) + 6 \times \big((-1) \times (-2) - 0 \times 0\big) + 4 \times \big((-1) \times 3 - 1 \times 0\big)$

$= 0 \times (-2) + 6 \times 2 + 4 \times (-3) = 0 + 12 - 12 = 0$

So the matrix is singular

69) c)

$$\begin{vmatrix} -1 & 2 & 3 \\ -2 & 3 & -5 \\ 1 & -2 & -2 \end{vmatrix} = (-1) \times \begin{vmatrix} 3 & -5 \\ -2 & -2 \end{vmatrix} - 2 \times \begin{vmatrix} -2 & -5 \\ 1 & -2 \end{vmatrix} + 3 \times \begin{vmatrix} -2 & 3 \\ 1 & -2 \end{vmatrix}$$

$$= (-1) \times \big(3 \times (-2) - (-5) \times (-2)\big) - 2 \times \big((-2) \times (-2) - (-5) \times 1\big) +$$

$$3 \times \big((-2) \times (-2) - 3 \times 1\big)$$

$$= (-1) \times (-16) - 2 \times 9 + 3 \times 1 = 16 - 18 + 3 = 1 \neq 0$$

So the matrix is not singular

69) d)

$$\begin{vmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 2\sqrt{8} & 1 \\ \sqrt{2} & -2\sqrt{2} & -\sqrt{8} \end{vmatrix} = \begin{vmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 4\sqrt{2} & 1 \\ \sqrt{2} & -2\sqrt{2} & -2\sqrt{2} \end{vmatrix}$$

$$= -\sqrt{2} \times \begin{vmatrix} 4\sqrt{2} & 1 \\ -2\sqrt{2} & -2\sqrt{2} \end{vmatrix} - 2\sqrt{2} \times \begin{vmatrix} -\sqrt{2} & 1 \\ \sqrt{2} & -2\sqrt{2} \end{vmatrix} + 2\sqrt{2} \times \begin{vmatrix} -\sqrt{2} & 4\sqrt{2} \\ \sqrt{2} & -2\sqrt{2} \end{vmatrix}$$

$$= -\sqrt{2} \times \Big(4\sqrt{2} \times \big(-2\sqrt{2}\big) - 1 \times \big(-2\sqrt{2}\big)\Big) - 2\sqrt{2} \times \Big(\big(-\sqrt{2}\big) \times \big(-2\sqrt{2}\big) - 1 \times \sqrt{2}\Big) +$$

$$2\sqrt{2} \times \Big(\big(-\sqrt{2}\big) \times \big(-2\sqrt{2}\big) - 4\sqrt{2} \times \sqrt{2}\Big)$$

$$= -\sqrt{2} \times \big(-16 + 2\sqrt{2}\big) - 2\sqrt{2} \times \big(4 - \sqrt{2}\big) + 2\sqrt{2} \times (-4)$$

$$= 16\sqrt{2} - 4 - 8\sqrt{2} + 4 - 8\sqrt{2} = 0$$

So the matrix is singular

70) a)

Calculator:

$$\det \mathbf{A} = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{vmatrix} = 6$$

By hand:

$$\det \mathbf{A} = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{vmatrix} = 1 \times \begin{vmatrix} 2 & 1 \\ 4 & -1 \end{vmatrix} - 1 \times \begin{vmatrix} 3 & 1 \\ 5 & -1 \end{vmatrix} + 2 \times \begin{vmatrix} 3 & 2 \\ 5 & 4 \end{vmatrix}$$

$$= 1 \times (2 \times (-1) - 1 \times 4) - 1 \times (3 \times (-1) - 1 \times 5) + 2 \times (3 \times 4 - 2 \times 5)$$

$$= 1 \times (-6) - 1 \times (-8) + 2 \times 2 = 6$$

$\therefore$ the matrix $\mathbf{A}$ is not singular

70) b)

Calculator:

$$\det \mathbf{B} = \begin{vmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{vmatrix} = 0$$

By hand:

$$\det \mathbf{B} = \begin{vmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{vmatrix} = -2 \times \begin{vmatrix} 1 & 5 \\ -2 & 1 \end{vmatrix} - 3 \times \begin{vmatrix} 4 & 5 \\ 3 & 1 \end{vmatrix} + 1 \times \begin{vmatrix} 4 & 1 \\ 3 & -2 \end{vmatrix}$$

$$= -2 \times \big(1 \times 1 - 5 \times (-2)\big) - 3 \times (4 \times 1 - 5 \times 3) + 1 \times (4 \times (-2) - 1 \times 3)$$

$$= -2 \times 11 - 3 \times (-11) + 1 \times (-11) = 0$$

$\therefore$ the matrix $\mathbf{B}$ is singular

70) c)

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = 6 \times 0 = 0$

$\therefore$ the matrix $\mathbf{AB}$ is singular

70) d)

$\det \mathbf{BA} = \det \mathbf{B} \times \det \mathbf{A} = 0 \times 6 = 0$

$\therefore$ the matrix $\mathbf{BA}$ is singular

71)

$$\begin{vmatrix} 2 & k & 3 \\ 2 & 2 & k \\ 0 & 1 & -1 \end{vmatrix} = 2 \times \begin{vmatrix} 2 & k \\ 1 & -1 \end{vmatrix} - k \times \begin{vmatrix} 2 & k \\ 0 & -1 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}$$

$$= 2 \times (2 \times (-1) - k \times 1) - k \times (2 \times (-1) - k \times 0) + 3 \times (2 \times 1 - 2 \times 0)$$

$$= 2 \times (-2 - k) - k \times (-2) + 3 \times 2 = -4 - 2k + 2k + 6 = 2 \neq 0$$

So the matrix $\mathbf{A}$ is never singular

72)

$$\begin{vmatrix} k^2 & 1 & 2 \\ -k^2 & 2k^2 & 3 \\ -1 & 1 & -2 \end{vmatrix} = k^2 \times \begin{vmatrix} 2k^2 & 3 \\ 1 & -2 \end{vmatrix} - 1 \times \begin{vmatrix} -k^2 & 3 \\ -1 & -2 \end{vmatrix} + 2 \times \begin{vmatrix} -k^2 & 2k^2 \\ -1 & 1 \end{vmatrix}$$

$$= k^2 \times (2k^2 \times (-2) - 3 \times 1) - 1 \times \big(-k^2 \times (-2) - 3 \times (-1)\big) + 2 \times \big(-k^2 \times 1 - 2k^2 \times (-1)\big)$$

$$= k^2 \times (-4k^2 - 3) - 1 \times (2k^2 + 3) + 2 \times (k^2) = -4k^4 - 3k^2 - 2k^2 - 3 + 2k^2$$

$$= -4k^4 - 3k^2 - 3$$

$k^4 \geq 0$ for all real values of k, so $-4k^4 \leq 0$ for all real values of k

$k^2 \geq 0$ for all real values of k, so $-3k^2 \leq 0$ for all real values of k

$\therefore -4k^4 - 3k^2 - 3 < 0$ for all real values of k, and so can never be equal to zero

So the matrix $\mathbf{A}$ is never singular

73)

$$\begin{vmatrix} -\sin\theta & 1 & \cos\theta \\ 0 & 1 & 0 \\ \cos\theta & 1 & \sin\theta \end{vmatrix} = -\sin\theta \times \begin{vmatrix} 1 & 0 \\ 1 & \sin\theta \end{vmatrix} - 1 \times \begin{vmatrix} 0 & 0 \\ \cos\theta & \sin\theta \end{vmatrix} + \cos\theta \times \begin{vmatrix} 0 & 1 \\ \cos\theta & 1 \end{vmatrix}$$

$= -\sin\theta \times (1 \times \sin\theta - 0 \times 1)$

$-1 \times (0 \times \sin\theta - 0 \times \cos\theta)$

$+ \cos\theta \times (0 \times 1 - 1 \times \cos\theta)$

$= -\sin\theta \times \sin\theta - 1 \times 0 + \cos\theta \times (-\cos\theta)$

$= -\sin^2\theta - \cos^2\theta = -(\sin^2\theta + \cos^2\theta) = -1 \neq 0$

So the matrix $\mathbf{A}$ is never singular

74)

$$\begin{vmatrix} \sin\theta - 1 & 1 & 1 - \cos\theta \\ 0 & 1 & 0 \\ 1 - \cos\theta & 1 & 1 - \sin\theta \end{vmatrix} = (\sin\theta - 1) \times \begin{vmatrix} 1 & 0 \\ 1 & 1 - \sin\theta \end{vmatrix} - 1 \times \begin{vmatrix} 0 & 0 \\ 1 - \cos\theta & 1 - \sin\theta \end{vmatrix} +$$

$$(1 - \cos\theta) \times \begin{vmatrix} 0 & 1 \\ 1 - \cos\theta & 1 \end{vmatrix}$$

$= (\sin\theta - 1) \times (1 \times (1 - \sin\theta) - 0 \times 1)$

$-1 \times \big(0 \times (1 - \sin\theta) - 0 \times (1 - \cos\theta)\big)$

$+(1 - \cos\theta) \times \big(0 \times 1 - 1 \times (1 - \cos\theta)\big)$

$= (\sin\theta - 1) \times (1 - \sin\theta) + (1 - \cos\theta) \times (\cos\theta - 1)$

$= 2\sin\theta - \sin^2\theta - 1 + 2\cos\theta - 1 - \cos^2\theta$

$= -(\sin^2\theta + \cos^2\theta) + 2\sin\theta + 2\cos\theta - 2$

$= -1 + 2\sin\theta + 2\cos\theta - 2$

$= 2\sin\theta + 2\cos\theta - 3$

$= 2\sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) - 3$

The maximum value of $2\sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) - 3$ is $2\sqrt{2} - 3 = -0.17157 \ldots < 0$

So the matrix $\mathbf{A}$ is never singular

75) a)

$$\begin{vmatrix} 7 & -1 & k \\ 2 & 1 & k \\ 0 & 5 & 4 \end{vmatrix} = 0 \Rightarrow 7 \times \begin{vmatrix} 1 & k \\ 5 & 4 \end{vmatrix} - (-1) \times \begin{vmatrix} 2 & k \\ 0 & 4 \end{vmatrix} + k \times \begin{vmatrix} 2 & 1 \\ 0 & 5 \end{vmatrix} = 0$$

$\Rightarrow 7 \times (1 \times 4 - k \times 5) - (-1) \times (2 \times 4 - k \times 0) + k \times (2 \times 5 - 1 \times 0) = 0$

$\Rightarrow 7 \times (4 - 5k) + 1 \times 8 + k \times 10 = 0$

$\Rightarrow 28 - 35k + 8 + 10k = 0 \Rightarrow 36 - 25k = 0 \Rightarrow k = \frac{36}{25}$

75) b)

$$\begin{vmatrix} -2 & k & 4 \\ -3 & 1 & k \\ 1 & 2 & 4 \end{vmatrix} = 0 \Rightarrow -2 \times \begin{vmatrix} 1 & k \\ 2 & 4 \end{vmatrix} - k \times \begin{vmatrix} -3 & k \\ 1 & 4 \end{vmatrix} + 4 \times \begin{vmatrix} -3 & 1 \\ 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow -2 \times (1 \times 4 - k \times 2) - k \times (-3 \times 4 - k \times 1) + 4 \times (-3 \times 2 - 1 \times 1) = 0$$

$$\Rightarrow -2 \times (4 - 2k) - k \times (-12 - k) + 4 \times (-7) = 0$$

$$\Rightarrow -8 + 4k + 12k + k^2 - 28 = 0 \Rightarrow k^2 + 16k - 36 = 0$$

$$\Rightarrow (k + 18)(k - 2) = 0 \Rightarrow k = -18 \text{ or } k = 2$$

75) c)

$$\begin{vmatrix} k+2 & 2 & -3 \\ 2 & -3 & 2 \\ 1 & 0 & k-2 \end{vmatrix} = 0$$

$$\Rightarrow (k+2) \times \begin{vmatrix} -3 & 2 \\ 0 & k-2 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 2 \\ 1 & k-2 \end{vmatrix} + (-3) \times \begin{vmatrix} 2 & -3 \\ 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (k+2) \times (-3 \times (k-2) - 2 \times 0)$$

$$-2 \times (2 \times (k-2) - 2 \times 1)$$

$$+(-3) \times (2 \times 0 - (-3) \times 1) = 0$$

$$\Rightarrow (k+2) \times (6 - 3k) - 2 \times (2k - 6) - 3 \times 3 = 0$$

$$\Rightarrow -3k^2 + 12 - 4k + 12 - 9 = 0 \Rightarrow 0 = 3k^2 + 4k - 15$$

$$\Rightarrow 0 = (3k - 5)(k + 3) \Rightarrow k = \frac{5}{3} \text{ or } k = -3$$

76)

A', B', C', D' and E' are coplanar if they lie in the same plane, so the image S' will have a volume of 0 cm^3. This will occur if $\det \mathbf{A} = 0$

$$\begin{vmatrix} 2k & 1 & k \\ 4k & 5 & 3 \\ 3 & -6 & k \end{vmatrix} = 0$$

$$\Rightarrow 2k \times \begin{vmatrix} 5 & 3 \\ -6 & k \end{vmatrix} - 1 \times \begin{vmatrix} 4k & 3 \\ 3 & k \end{vmatrix} + k \times \begin{vmatrix} 4k & 5 \\ 3 & -6 \end{vmatrix} = 0$$

$$\Rightarrow 2k \times \big(5 \times k - 3 \times (-6)\big) - 1 \times (4k \times k - 3 \times 3) + k \times (4k \times (-6) - 5 \times 3) = 0$$

$$\Rightarrow 2k \times (5k + 18) - 1 \times (4k^2 - 9) + k \times (-24k - 15) = 0$$

$$\Rightarrow 10k^2 + 36k - 4k^2 + 9 - 24k^2 - 15k = 0 \Rightarrow 0 = 18k^2 - 21k - 9$$

$$\Rightarrow 0 = 6k^2 - 7k - 3 \Rightarrow 0 = (3k + 1)(2k - 3) \Rightarrow k = -\frac{1}{3} \text{ or } k = \frac{3}{2}$$

77)

Let A and B be two square matrices.

If $\det \mathbf{A} = 0$, then $\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = 0 \times \det \mathbf{B} = 0$

If $\det \mathbf{B} = 0$, then $\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = \det \mathbf{A} \times 0 = 0$

If $\det \mathbf{A} = 0$ *and* $\det \mathbf{B} = 0$, then $\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = 0 \times 0 = 0$

∴ if at least one of two square matrices are singular, then their product will be a singular matrix.

78)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\det \mathbf{A} = ad - bc \neq 0$

Let $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then $\det \mathbf{B} = ps - qr \neq 0$

$\det \mathbf{A} + \det \mathbf{B} = ad - bc + ps - qr$

$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} a+p & b+q \\ c+r & d+s \end{pmatrix}$

$\det(\mathbf{A} + \mathbf{B}) = (a+p) \times (d+s) - (b+q) \times (c+r)$

$= ad + as + pd + ps - bc - br - qc - qr$

$= (ad - bc + ps - qr) + (as + pd - br - qc)$

$ad - bc + ps - qr \neq (ad - bc + ps - qr) + (as + pd - br - qc)$

∴ $\det(\mathbf{A} + \mathbf{B}) \neq \det \mathbf{A} + \det \mathbf{B}$

79)

Let $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$, then $\det \mathbf{A} = 1 \neq 0$

Let $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$, then $\det \mathbf{B} = -1 \neq 0$

$\det \mathbf{A} + \det \mathbf{B} = 1 + (-1) = 0$

$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix}$

$\det(\mathbf{A} + \mathbf{B}) = 2 \times 0 - 2 \times 0 = 0$

$\therefore \det(\mathbf{A} + \mathbf{B}) = \det \mathbf{A} + \det \mathbf{B}$ in this case

Alternative Method:

Consider $\mathbf{A}$ and $\mathbf{B}$ as rotations about the origin by $45°$, with one being anticlockwise and the other being clockwise.

Let $\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, then $\det \mathbf{A} = 1 \neq 0$

Let $\mathbf{B} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$, then $\det \mathbf{B} = 1 \neq 0$

$\det \mathbf{A} + \det \mathbf{B} = 1 + 1 = 2$

$\mathbf{A} + \mathbf{B} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} + \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{2}} & 0 \\ 0 & \frac{2}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$\det(\mathbf{A} + \mathbf{B}) = \sqrt{2} \times \sqrt{2} - 0 \times 0 = 2$

$\therefore \det(\mathbf{A} + \mathbf{B}) = \det \mathbf{A} + \det \mathbf{B}$ in this case

80)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then $\det \mathbf{A} = ad - bc \neq 0$

Let $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then $\det \mathbf{B} = ps - qr \neq 0$

$\det \mathbf{A} \times \det \mathbf{B} = (ad - bc)(ps - qr) = adps - adqr - bcps + bcqr$

$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$

$\det \mathbf{AB} = (ap + br) \times (cq + ds) - (aq + bs) \times (cp + dr)$

$= apcq + apds + brcq + brds - aqcp - aqdr - bscp - bsdr$

$= acpq + adps + bcqr + bdrs - acpq - adqr - bcps - bdrs$

$= adps + bcqr - adqr - bcps$

$= adps - adqr - bcps + bcqr$

$\therefore \det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B}$